

Sprint 1 Requirements Specification

Live Product Foundation

AI-Native Real Estate Platform — Agile MVP

Sub-document of: AI-Native Real Estate Master PRD v2.0 (Agile MVP)

What this document is

This is a sprint-level breakdown built from Section 15.5 (Initial Sprint Plan) and Section 16 (Requirements-to-Sprint Traceability) of the Master PRD. It takes the Sprint 1 increment — a responsive web shell, a versioned API, CI/CD, staging, authentication with roles and consent, buyer/seller dashboards, and an audit skeleton — and turns it into requirements the team can actually plan, build, and demo against. It doesn't replace the Master PRD; wherever this document doesn't say something explicitly, the Master PRD is still the final word.

Document Control

Field	Value
Parent document	AI-Native Real Estate Master PRD v2.0 (Agile MVP)
Sprint	Sprint 1 of 10 — "Live product foundation"
Sprint length	Two weeks (per Master PRD Section 15.1)
Primary requirements covered	REQ-01 (Authentication, Roles and Consent); REQ-16 (Mobile Portability — introduced this sprint)
Cross-cutting NFRs introduced	Security/RBAC, Auditability, Mobile Portability & Accessibility, Observability (started here, continuing into Sprint 2), Performance/Reliability/Idempotency, Legal/Compliance Configuration (also spanning Sprint 1-2)
Status	Draft — for sprint planning

1. Purpose and Scope

This document takes the Sprint 1 row of the Master PRD's Initial Sprint Plan (Section 15.5) and turns it into requirements the team can build against. Sprint 1 is the foundation sprint — it's where we stand up the live, deployable shell of the product that every sprint after this one (2 through 10) builds on top of.

Under the Master PRD's Iterative Scope Rule (Section 15), this plan should be read as an initial forecast rather than a locked waterfall schedule. Stories can be split or reshaped after the sprint demo based on what we learn. The one hard rule: if a PO requirement needs to change, that change has to be reflected in a newly approved PRD version before we drop anything.

1.1 Sprint 1 Goal (from Master PRD Section 15.5)

Field	Master PRD content
Sprint / goal	Sprint 1 — Live product foundation
Increment scope	Responsive web shell; versioned API; CI/CD; staging; auth, roles, consent; buyer/seller dashboards; audit skeleton.
Mandatory usable demo	Register as a buyer and as a seller on both desktop and mobile, verify email, log in and out, and reach the correct role dashboard — all on live

	staging.
Primary feedback sought	How usable the onboarding feels, and whether the API/mobile foundation holds up.

2. Governance Carried Over from the Master PRD

Every story pulled into Sprint 1 has to follow the same rules laid out in Master PRD Sections 15.2 through 15.4. Nothing changes here — this is just a reminder of what "ready" and "done" mean going into this sprint.

2.1 Definition of Ready

- The user, the outcome, and the business reason behind the story are all clear.
- Acceptance criteria and the mandatory demo path are written down.
- Data, legal/policy, and dependency assumptions have been identified.
- AI autonomy level, confidence policy, fallback behaviour, and audit requirements are defined.
- Mobile-browser behaviour and API impact are understood before work starts.
- The story is small enough to complete, integrate, and demo within the sprint.

2.2 Definition of Done

- Built end to end in the shared codebase — no "we'll wire this up later" work left on the demonstrated path.
- Automated unit, integration, API contract, security, and relevant AI evaluation tests all pass.
- Architecture boundaries, audit events, observability, and feature flags are in place.
- The responsive desktop and mobile-browser flow has been tested.
- Acceptance criteria are demonstrated live, using a real role and real persisted data — not a mock.
- Demo feedback and defects are captured, and documentation/backlog are updated accordingly.
- No unresolved critical security, privacy, data-integrity, or confidentiality defect remains.

2.3 Mandatory Sprint Demo Standard

Every sprint demo, including this one, has to follow the same script:

- Start from the current deployed staging product — not a local prototype.
- Use an end-user role and complete a task that actually means something; if the work is operations-only, show an observable operational capability instead.
- Show persisted state, and show how it affects the next stage of the workflow.
- Show at least one happy path and one exception/failure path for whatever new capability is being demoed.
- Show the new flow on desktop and on at least one mobile-browser viewport.
- Show the relevant audit/AI-decision evidence for anything autonomous.
- Capture stakeholder observations as backlog items before the review closes.
- If the integrated demo can't actually run, the increment isn't accepted as Done — no exceptions.

3. Functional Requirements — Sprint 1

Each requirement below traces back to a Sprint 1 increment item in Section 15.5, filled out using the matching detailed requirement, architecture module, and ADR from the Master PRD (Sections 6, 8, and 14).

FR-S1-01 — Account Registration, Login and Verification (from REQ-01)

Source: Master PRD Section 6, REQ-01 — Authentication, Roles and Consent.

What it covers: users need to be able to register and log in using either email or Google, verify their email, optionally verify a phone number, reset a forgotten password, and log out cleanly.

AI-native behaviour: the AI greets users and routes them by role, explains consent in plain language rather than legal boilerplate, notices when onboarding has been left half-finished and nudges the user toward the next step. If something looks risky, the system can quietly ask for step-up verification without exposing its internal reasoning.

User control and boundaries: users are in control of their own consent and can withdraw anything non-essential. Admin actions should never surface to a regular user, under any circumstance.

Acceptance criteria: a new buyer or seller should be able to create an account, verify their identity channels, give consent, log in and out, and land on the correct responsive dashboard — all within five minutes, all through versioned APIs, and all passing mobile-browser tests.

FR-S1-02 — Role Selection and Buyer/Seller Dashboards (from REQ-01)

Source: Master PRD Section 6 (REQ-01) and the "buyer/seller dashboards" line item in Section 15.5.

What it covers: at onboarding, users choose whether they're a Buyer or a Seller and get routed to the dashboard built for that role. Admin and exception-handling users are never created through self-service — only through controlled internal provisioning.

User control and boundaries: admin actions and admin-only screens must stay completely invisible to normal Buyer/Seller users.

Acceptance criteria: a registered user reaches their correct role dashboard, on both desktop and mobile-browser viewports, on live staging.

FR-S1-03 — Consent Capture and Management (from REQ-01)

Source: Master PRD Section 6, REQ-01.

What it covers: at onboarding, we capture consent records for AI use, verification processing, communications, and document sharing.

AI-native behaviour: before capturing consent, the AI explains each item in plain language so the user actually understands what they're agreeing to.

User control and boundaries: users can withdraw consent for anything non-essential after their account is created.

Acceptance criteria: consent is recorded per category, visible to the user, and withdrawable where non-essential — and all of that needs to check out against the persisted state, not just the UI.

FR-S1-04 — Responsive Web Shell and Mobile-Browser Support (from REQ-16)

Source: Master PRD Section 6, REQ-16 — Mobile-Portability and Client-Neutral Product Design (introduced this sprint); Section 14 (Mobile Portability and Architecture Evolution).

What it covers: the MVP ships as a responsive website, but every Sprint 1 capability — registration, login, role dashboards — needs to be reachable through versioned, role-secured, task-specific APIs. Nothing should depend on web-session-only logic.

AI-native behaviour: onboarding and AI guidance run as task APIs with resumable state, not as a chat session tied to one browser tab.

User control and boundaries: the web presentation can vary however it needs to, but no business rule or source-of-truth state is allowed to live only in the browser.

Acceptance criteria: all Sprint 1 flows pass both desktop and mobile-browser tests, and registration, login, and dashboard retrieval are demonstrably callable through the same versioned APIs the web client itself uses.

FR-S1-05 — Versioned, API-First Foundation

Source: the "versioned API" line in Master PRD Section 15.5; ADR-016 (API-first, client-neutral business logic); Section 10.3 (API and client-contract principles).

What it covers: every Sprint 1 capability sits behind a versioned API contract owned by the Auth & Identity module, independent of whatever the web presentation layer looks like.

Why it matters: this is what lets us pull out a native client later (Section 14) without having to rework the backend.

Acceptance criteria: every Sprint 1 endpoint carries an API version, and a client-neutral request can complete authentication without ever touching the web UI.

FR-S1-06 — CI/CD Pipeline and Shared Staging Environment

Source: the "CI/CD; staging" line in Master PRD Section 15.5; Section 15.3 Definition of Done; Section 15.4 demo standard.

What it covers: an automated CI/CD pipeline builds, tests, and deploys the product to a shared staging environment, and that's the environment every sprint demo runs from — per Section 15.4's rule to "start from the current deployed staging product, not a local prototype."

Acceptance criteria: a change merged to mainline gets automatically built, tested, and deployed to staging, and the Sprint 1 demo runs from that deployed environment.

FR-S1-07 — Audit Skeleton (foundational)

Source: the "audit skeleton" line in Master PRD Section 15.5; Section 8.1's "Audit & Observability" bounded context (owns append-only audit records, traces, metrics, and quality samples); Section 16.1's cross-cutting NFR "Auditability and immutable events," introduced this sprint.

What it covers: we stand up a minimal, append-only audit capability so that Auth & Identity events — registration, login, logout, consent capture/withdrawal, role selection — are recorded starting in Sprint 1, ahead of the heavier transactional and AI-decision audit needs that show up in later sprints.

Acceptance criteria: for a demo user, the registration, login/logout, consent, and role-selection events all show up in an append-only audit log tied to that user's account.

4. Non-Functional Requirements — Sprint 1

Master PRD Section 16.1 (Cross-cutting NFR Traceability) lays out which NFR areas need to show up starting in Sprint 1 versus which ones get demonstrated continuously afterward. The table below pulls in the full NFR definitions from Section 7, filtered down to whatever's flagged as "Sprint 1" or "Sprint 1-2."

NFR area	Introduced	Sprint 1 requirement (from Master PRD Section 7)
Security	Sprint 1	RBAC/ABAC enforced on every endpoint, least-privilege access, secure file validation, encrypted transport and storage, proper secret management, rate limiting, and vulnerability scanning.
Privacy	Sprint 1	Data minimisation, purpose limitation, a clear public/private separation, consent records, role-scoped document access, and configurable retention/deletion.
Auditability	Sprint 1	Auth & Identity events (registration, login, consent, role selection) are immutable or versioned from Sprint 1 onward; full transactional/AI-decision auditability grows with each sprint after this.
Performance	Sprint 1	Common page and API responses should hit a p95 under two seconds.
Reliability	Sprint 1	Critical status transitions — registration, role assignment — are transactional, idempotent, and exactly-once in effect.
Availability	Sprint 1	Pilot target of 99.5% monthly availability for core flows; login and dashboard access need to stay up even if other, non-Sprint-1 features degrade.
Observability	Sprint 1-2	Central logs, metrics, traces, and alerts get stood up for the Sprint 1 shell; AI cost and decision-quality observability expands starting Sprint 2.
Mobile portability	Sprint 1	Responsive layouts, touch-safe controls, API contract tests, resumable workflows, and no business logic that only lives on the web.
Accessibility	Sprint 1	Core flows — registration, login, dashboards — meet WCAG-oriented requirements for labels, keyboard navigation, focus, contrast, error handling, and screen-reader semantics.
Maintainability	Sprint 1	Bounded modules, typed API contracts, automated tests, architecture checks, code review, and ADRs — applied to AI-generated code just as much as anything else, starting from this first increment.
Legal / compliance configuration	Sprint 1-2	No legal advice, escrow, title transfer, payment custody, mortgage execution, or binding agreement generation. For Sprint 1 specifically, this mostly means onboarding and consent language must not imply any of these.

4.1 Architecture Constraints Applicable to Sprint 1

- The Auth & Identity module owns accounts, roles, sessions, consent, and authentication state — it must not own listings, pricing, offers, or verification-evidence decisions (Section 8.1).
- The Audit & Observability module owns append-only audit records, traces, metrics, and quality samples — it must not own operational business decisions (Section 8.1).
- ADR-001/002: build on a microservice-ready modular monolith; don't start with full microservices.
- ADR-008: use internal domain events inside the monolith to support audit, notifications, debugging, and later extraction.
- ADR-016: keep business logic API-first and client-neutral, so mobile portability doesn't require backend rework down the line.
- ADR-018: Sprint 1 needs to produce a live, usable, integrated increment — not slides, wireframes, or a set of APIs sitting in isolation.

5. Sprint 1 Acceptance Criteria and Demo Script Outline

Pulled together from the Sprint 1 row of Section 15.5 and the acceptance criteria under REQ-01 and REQ-16 (Section 6).

- Register as a new Buyer on desktop, verify email, log in, and reach the Buyer dashboard.
- Register as a new Seller on a mobile-browser viewport, verify email, log in, and reach the Seller dashboard.
- Demonstrate password reset and logout for at least one role.
- Demonstrate a consent capture and a consent withdrawal for a non-essential processing category.
- Show the same registration, login, and dashboard-retrieval actions succeeding through the versioned API, independent of the web UI — the client-neutral proof required by REQ-16.
- Show the audit log entries created by the demoed registration, login, and consent events.
- Show one exception/failure path — an invalid verification code, or a failed login attempt — alongside the happy path, per the mandatory demo standard.
- Confirm the whole demo runs from the shared staging environment deployed via CI/CD, not from a local build.

6. Sprint 1 Requirements Traceability

Pulled from Master PRD Section 16 (Requirements-to-Sprint Traceability) and Section 16.1 (Cross-cutting NFR Traceability), filtered down to whatever's primarily introduced in Sprint 1.

Requirement / concern	Primary / introduced sprint	Continuous integration	Sprint 1 traceability evidence
REQ-01 — Authentication, Roles and Consent	1	2–10	Registration, consent, and role dashboards demonstrated on live staging.
REQ-16 — Mobile Portability	1	All	Mobile-browser demo, plus independent-client proof for auth and dashboard retrieval.
Security, privacy and RBAC	1	Every sprint; full review at Sprint 10	RBAC/ABAC enforced on Sprint 1 auth/dashboard endpoints.
Auditability and immutable events	1	Every autonomous/transactional	Audit skeleton capturing Sprint 1 identity events.

Requirement / concern	Primary / introduced sprint	Continuous integration	Sprint 1 traceability evidence
		demo	
Mobile portability and accessibility	1	Mobile-browser demo each sprint; full gate at Sprint 10	Responsive shell tested on desktop and mobile-browser.
Observability and cost/quality metrics	1-2	Expanded with each new capability	Logs, metrics, and traces stood up for the Sprint 1 shell.
Performance, reliability and idempotency	1	Story acceptance and hardening through Sprint 10	p95 response targets and idempotent registration/login.
Legal/compliance configuration	1-2	Feature-specific gates; final approval at Sprint 10	Onboarding/consent copy avoids implying legal advice, escrow, or binding agreements.

7. Explicitly Out of Scope for Sprint 1

The following requirements from Master PRD Section 6 aren't part of Sprint 1. Each picks up at its own designated sprint per Master PRD Section 16:

- REQ-02 Autonomous Seller and Property Verification — Sprint 2
- REQ-03 AI-Guided Listing Creation — Sprint 3
- REQ-04 Media and Document Intelligence — Sprints 2-3
- REQ-05 Autonomous Listing Approval — Sprint 3
- REQ-06 Buyer Search and Recommendations — Sprint 5
- REQ-07 Property Details and AI Explainer — Sprint 5
- REQ-08 Enquiries, Messaging and Moderation — Sprint 6
- REQ-09 Visit Scheduling — Sprint 6
- REQ-10 Preliminary Offer and Negotiation — Sprint 7
- REQ-11 Checklist and Lawyer Handoff — Sprint 9
- REQ-12 Autonomous Operations / Exceptions — Sprint 2
- REQ-13 Notifications — Sprint 2
- REQ-14 Pricing Recommendation — Sprint 4
- REQ-15 Proxy Bidding — Sprint 8
- REQ-17 AI Governance and Evaluation (as a full capability) — Sprint 2

Note on scope changes

Under the Master PRD's Iterative Scope Rule (Section 15), this Sprint 1 breakdown may still be split or reprioritised after the sprint demo. That said, any proposed removal or material change to a P0 requirement (REQ-01 or REQ-16) has to be reflected in a newly approved Master PRD version first.